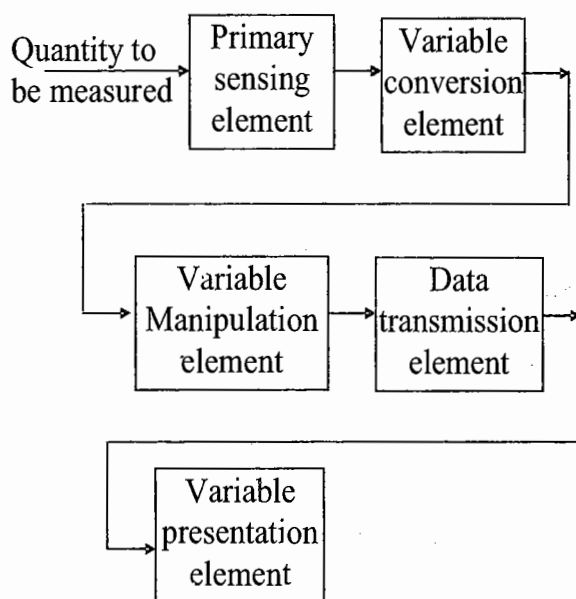


# 1

# Basics of Transducers

It is important to have a systematic organization and analysis of measurement systems. An instrument may be defined as a device or a system which is designed to maintain a functional relationship between prescribed properties of physical variables and must include ways and means of communication to a human observer.

## Generalized Measurement system:



### Primary sensing element :

The quantity under measurement makes its first contact with the primary sensing element of a measurement system.

### Variable conversion element:

The output of the primary sensing element is converted to some other suitable form for the instrument to perform desired function.

### Variable manipulation Element :

The function of this element is to manipulate the signal presented to it preserving the original nature of the signal.

### Data Transmission Element:

This element transmits the data from one to another.

### Data presentation element:

This element conveys the information about the quantity under measurement to the personnel handling the instrument or the system for monitoring, control & analysis purposes.

- The measurand in an instrumentation system makes contact with a primary detection element or input device.
- The measurand or the input signal is called an “information” for measurement system.
- The information may be in the form of a physical phenomenon or it may be an electrical signal.
- The process of detection and conversion of the information into an acceptable form requires energy.
- The ideal situation is, the energy required for above purpose is supplied from outside, not from measurand.
- So that the measurand is not distorted and the analogous output of the detection is a faithful representation of measurand.

## Transducers

An electronic instrumentation system consists of a number of components to perform a measurement and record its results.

A generalized measurement system consists of three major components.

- An input device
- A signal conditioning or processing device
- An output device
- The input device receives the measurand or the quantity under measurement and delivers a proportional or analogous electrical signal to the signal conditioning device.



- Here the signal is amplified, attenuated, filtered, modulated or otherwise modified in format acceptable to the output device.

**Note:** The input quantity for the most instrumentation systems is a “Non – electrical quantity”. In order to use electrical methods and techniques for measurement, manipulation or control, non – electrical quantity is generally converted into an electrical form by a device called a “transducer”.

- We can define transducer as a device which, when accurately transforms energy from one form to another.
- Another name for transducer is “pick – up”.
- The reasons for transforming a physical phenomenon into an electrical form is that the electrical output can be easily used, transmitted and processed for the purpose of measurement. Modern digital computers make the use of these transducers absolutely essential.
- The relationship between the physical parameter and its resulting electrical signal must be linear.
- The operating range of the transducer is called the dynamic range and should be wide.

**Transducer mainly consists of two parts**

- Sensing element
- Transduction element

**Sensing element:**

A detector or sensing element is that part of a transducer which responds to a physical phenomenon or a change in a physical phenomenon.

**Transduction element:**

A transduction element transforms the output of a sensing element to an electrical output.

### **Classification of Transducers**

The transducers can be classified

- Based upon transduction principle
- as primary and secondary transducers
- as passive and active transducers

- as analog and digital transducers
- as transducers and inverse transducers

### **Based upon transduction principle**

The transducers can be classified on the basis of principle of transduction as resistive, inductive, capacitive etc., depending upon how they convert the input quantity into resistance, inductance or capacitance respectively.

### **Primary and Secondary Transducers:**

The first transducer which converts physical phenomenon into displacement, pressure, velocity etc. which is to be accepted by next stage is known as “Primary Transducer”.

The output of the primary transducer is converted subsequently into a usable output by a device called “Secondary Transducer”

### **Passive and Active Transducers:**

**Passive transducers:** They derive the power required for transduction from an auxiliary power source.

**Eg:** Resistive, inductive and capacitive transducers.

**Active transducers:** They do not require an auxiliary power source to produce their output. They are also known as self – generating type since they develop their own voltage or current output.

**Eg:** piezoelectric, photovoltaic etc.

### **Analog and digital Transducers:**

**Analog transducers:** These transducers convert the input quantity into an analog output which is a continuous function of time.

**Eg:** LVDT, thermocouple etc.

**Digital Transducers:-**These transducers convert the input quantity into an electrical output which is in the form of pulses.

### **Transducers & Inverse Transducers**

**Transducer:**A transducer can be broadly defined as a device which converts a non – electrical quantity into an electrical quantity.

**Inverse transducer:** An inverse transducer is defined as a device which converts an electrical quantity into a non – electrical quantity.



## Mechanical Devices as Primary Detectors

| Type                                        | Operation                        |
|---------------------------------------------|----------------------------------|
| <b>A. Contacting spindle, pin or finger</b> | Displacement to displacement     |
| <b>B. Elastic member</b>                    |                                  |
| 1. Prooving ring                            | Force to displacement            |
| 2. Bourdon tube                             | Pressure to displacement         |
| 3. Bellows                                  | Pressure to displacement         |
| 4. Diaphragm                                | Pressure to displacement         |
| 5. Spring                                   | Force to displacement            |
| <b>C. Mass</b>                              |                                  |
| 1. Seismic mass                             | Forcing function to displacement |
| 2. Pendulum scale                           | Force to displacement            |
| 3. Manometer                                | Pressure to displacement         |
| <b>D. Thermal</b>                           |                                  |
| 1. Thermocouple                             | Temperature to electric current  |
| 2. Bimetallic                               | Temperature to displacement      |
| 3. Temp – stick                             | Temperature to phase             |
| <b>E. Hydro pneumatic</b>                   |                                  |
| <b>1. Static</b>                            |                                  |
| (a) Float                                   | Fluid level to displacement      |
| (b) Hydrometer                              | Specific gravity to displacement |
| <b>2. Dynamic</b>                           |                                  |
| (a) Orifice                                 | Velocity to pressure             |
| (b) Venturi                                 | Velocity to pressure             |
| (c) Pitot tube                              | Velocity to pressure             |
| (d) Vanes                                   | Velocity to force                |
| (e) Turbines                                | Linear to angular velocity       |



## *Types of Electrical Transducers*

### *Passive Transducers*

| Electrical parameter And class of transducer | Principle of Operation                                                                                             | Typical applications        |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Potentiometer device                         | Positioning of the slider by an external force varies the resistance in a potentiometer or a bridge circuit.       | Pressure, displacement      |
| Resistance strain gauge                      | Resistance of a wire or semiconductor is changed by elongation or compression due to externally applied stress.    | Force, torque, displacement |
| Pirani gauge or hot wire meter               | Resistance of a heating element is varied by convection cooling of a stream of gas.                                | Gas flow, gas pressure      |
| Resistance thermometer                       | Resistance of pure metal wire with a large positive temperature co-efficient of resistance varies with temperature | Temperature, radiant heat   |
| Thermistor                                   | Resistance of certain metal oxides with negative temperature coefficient of resistance varies with temperature.    | Temperature, flow           |
| Resistance Hydrometer                        | Resistance of a conductive strip changes with moisture content.                                                    | Relative humidity           |
| Photoconductive cell                         | Resistance of the cell as a circuit element varies with incident light.                                            | Photo-sensitive relay.      |



## Capacitance

|                                     |                                                                                      |                         |
|-------------------------------------|--------------------------------------------------------------------------------------|-------------------------|
| Variable capacitance pressure gauge | Distance between two parallel plates is varied by an externally applied force.       | Displacement, pressure  |
| Capacitor microphone                | Sound pressure varies the capacitance between a fixed plate and a movable diaphragm. | Speech, music, noise    |
| Dielectric gauge                    | Variation in capacitance by changes in the dielectric or dielectric constant.        | Liquid level, thickness |

## Inductance

|                             |                                                                                                                                                     |                                              |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Magnetic circuit transducer | Self – inductance or mutual inductance of a.c. excited coil is varied by changes in the magnetic circuit.                                           | Pressure, displacement                       |
| Reluctance pick up          | Reluctance of the magnetic circuits is varied by changing the position of the iron core of coil.                                                    | Pressure, displacement, vibrations, position |
| Differential transformer    | The differential voltage of two secondary windings of a transformer is varied by positioning the magnetic core through an externally applied force. | Pressure, force, displacement, position.     |
| Eddy current gauge          | Inductance of a coil is varied by the proximity of an eddy current plate                                                                            | Displacement, thickness                      |
| Magnetostriction gauge      | Magnetic properties are varied by pressure and stress                                                                                               | Force, pressure, sound.                      |



## Voltage and Current

|                      |                                                                                                                                    |                                             |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Hall effect pick up  | A potential difference is generated across a semiconductor plate (germanium) when magnetic flux interacts with an applied current. | Magnetic flux, current, power               |
| Ionization chamber   | Electron flow induced by ionization of gas due to radio – active radiation.                                                        | Particle counting, radiation                |
| Photo emissive cell  | Electron emission due to incident radiation upon photo emissive surface.                                                           | Light and radiation.                        |
| Photomultiplier tube | Secondary electron emission due to incident radiation on photosensitive cathode                                                    | Light and radiation, Photosensitive relays. |

## Self – Generating Transducers

|                             |                                                                                                                  |                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Thermocouple and thermopile | An emf is generated across the junction of two dissimilar metals or semiconductors when that junction is heated. | Temperature, heat flow, radiation                  |
| Moving coil generator       | Motion of coil in a magnetic field generates a voltage.                                                          | Velocity, vibrations.                              |
| Piezoelectric pickup        | An emf is generated when an external force is applied to certain crystalline materials such as quartz.           | Sound, vibrations, acceleration, pressure changes. |
| Photovoltaic                | A voltage is generated in a semiconductor junction device when radiant energy stimulates the cell                | Light meter, solar cell                            |



### **Characteristics & Choice of Transducers**

When choosing a transducer for any application the input, transfer and output characteristics have to be taken into account.

#### **Input characteristics:**

The following are the input characteristics to be taken into consideration.

- Type of input and operating range
- Loading effects

#### **Transfer Characteristics:**

The transfer characteristics of transducers require attention of three separate elements.

- Transfer function
- Error
- Response of transducer to environmental influences.

#### **Output Characteristics:**

The three conditions in the output characteristics which should be considered are

- Type of electrical output
- Output impedance
- Useful range

#### **Factors influencing the choice of transducers:**

The following are the factors influencing the choice of transducer for measurement of a physical quantity:

- Operating principle
- Sensitivity
- Operating range
- Accuracy

- Cross sensitivity
- Errors
- Transient and Frequency response
- Loading effects
- Environmental compatibility
- Insensitivity to unwanted signals
- Usage and Ruggedness
- Electrical aspects
- Stability and reliability
- Static characteristics

#### **Example 01**

Which one of the following is an active transducer?

- (a) Photo voltaic cell
- (b) Strain gauge
- (c) Photo emissive cell
- (d) Synchronous

**Ans: (a)**

**Sol:** photo voltaic cell is an active transducer

#### **Example 02:**

A transducer has an output impedance of  $1 \text{ k}\Omega$  and the load resistance is  $1 \text{ M}\Omega$ , the transducer behaves as

- (a) a constant current source
- (b) a constant voltage source
- (c) a constant power source
- (d) none of the above

**Ans: (b)**